

Quantum Physics Pset 3
Katherine Choi

1. Calculate $\langle x \rangle, \langle p \rangle, \langle x^2 \rangle, \langle p^2 \rangle, \langle T \rangle$

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-)$$

$$\hat{p} = i\sqrt{\frac{\hbar m\omega}{2}} (\hat{a}_+ - \hat{a}_-)$$

$$\hat{a}_+ \psi_n = \sqrt{n+1} \psi_{n+1}$$

$$\hat{a}_- \psi_n = \sqrt{n} \psi_{n-1}$$

$$\begin{aligned} \langle x \rangle &= \sqrt{\frac{\hbar}{2m\omega}} \langle \psi_n | \hat{a}_+ + \hat{a}_- | \psi_n \rangle \\ &= \sqrt{\frac{\hbar}{2m\omega}} (\underbrace{\sqrt{n+1} \langle \psi_n | \psi_{n+1} \rangle}_{=0} + \underbrace{\sqrt{n} \langle \psi_n | \psi_{n-1} \rangle}_{\text{orthogonal} = 0}) \end{aligned}$$

$$\therefore \langle x \rangle = 0$$

$$\langle p \rangle = i\sqrt{\frac{\hbar m\omega}{2}} (\underbrace{\sqrt{n+1} \langle \psi_n | \psi_{n+1} \rangle}_{=0} - \underbrace{\sqrt{n} \langle \psi_n | \psi_{n-1} \rangle}_{=0})$$

$$\therefore \langle p \rangle = 0$$

$$\hat{x}^2 = \frac{\hbar}{2m\omega} (\hat{a}_+ + \hat{a}_-) (\hat{a}_+ + \hat{a}_-) = \frac{\hbar}{2m\omega} (\cancel{\hat{a}_+^2} + \hat{a}_+ \hat{a}_- + \hat{a}_- \hat{a}_+ + \cancel{\hat{a}_-^2})$$

$$\hat{a}_+ \hat{a}_- \psi_n = n \psi_n$$

$$\hat{a}_- \hat{a}_+ \psi_n = (n+1) \psi_n$$

$$\Rightarrow \langle x^2 \rangle = \frac{\hbar}{2m\omega} (n + n+1)$$

$$\therefore \langle x^2 \rangle = \frac{\hbar}{2m\omega} (2n+1)$$

$$\hat{p}^2 = -\frac{\hbar m\omega}{2} (\hat{a}_+ - \hat{a}_-) (\hat{a}_+ - \hat{a}_-) = -\frac{\hbar m\omega}{2} (\hat{a}_+^2 - \hat{a}_+ \hat{a}_- - \hat{a}_- \hat{a}_+ + \hat{a}_-^2)$$

$$= \frac{\hbar m\omega}{2} (n + n+1)$$

$$\therefore \langle p^2 \rangle = \frac{\hbar m\omega}{2} (2n+1)$$

$$\langle T \rangle = \frac{\langle p^2 \rangle}{2m} = \frac{\hbar m \omega (n + 1/2)}{2m} = \frac{\hbar \omega}{2} (n + \frac{1}{2})$$

uncertainty principle

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\frac{\hbar}{2m\omega} (2n+1)}$$

$$\sigma_p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{\frac{\hbar m \omega}{2} (2n+1)}$$

$$\sigma_x \sigma_p = \sqrt{\frac{\hbar}{2m\omega} (2n+1) \cdot \frac{\hbar m \omega}{2} (2n+1)} = \hbar (n + \frac{1}{2})$$

$$\text{Since } n + \frac{1}{2} \geq \frac{1}{2}, \quad \sigma_x \sigma_p \geq \frac{\hbar}{2}$$

So the uncertainty principle is satisfied.

2. $\Psi(x, 0) = A[3\psi_1(x) - 4i\psi_2(x)]$

(a) find A

$$\int |\Psi|^2 dx = 1 \Rightarrow \int |A|^2 |3\psi_1 - 4i\psi_2|^2 dx = 1$$

$$1 = |A|^2 (9 \langle \psi_1 | \psi_1 \rangle + 16 \langle \psi_2 | \psi_2 \rangle)$$

$$|A|^2 (9+16) = 25 |A|^2 = 1 \rightarrow A = \frac{1}{5}$$

(b) construct $\Psi(x, t)$, $|\Psi(x, t)|^2$, oscillation frequency.

$$\psi_n(x, t) = \psi_n(x) e^{-iE_n t / \hbar}$$

$$\Psi(x, t) = \frac{1}{5} (3\psi_1(x) e^{-iE_1 t / \hbar} - 4i\psi_2(x) e^{-iE_2 t / \hbar})$$

$$\begin{aligned} |\Psi|^2 &= \Psi \Psi^* = \frac{1}{25} (3\psi_1 e^{iE_1 t / \hbar} + 4i\psi_2 e^{E_2 t / \hbar}) (3\psi_1 e^{-iE_1 t / \hbar} - 4i\psi_2 e^{-iE_2 t / \hbar}) \\ &= \frac{1}{25} (9\psi_1^2 + 16\psi_2^2 - 12i\psi_1\psi_2 e^{-i(E_2 - E_1)t / \hbar} + 12i\psi_1\psi_2 e^{i(E_2 - E_1)t / \hbar}) \\ &= -24\psi_1\psi_2 \sin(\omega t) \end{aligned}$$

oscillation frequency: ω .

ψ_3 instead of ψ_2 will give us $E_3 - E_1 = (3.5 - 1.5)\hbar\omega = 2\hbar\omega$.

So the frequency would be 2ω .

(c) calculate $\langle x \rangle, \langle p \rangle$.

$$\langle x \rangle_t = \int \psi^* x \psi dx$$

ψ_1, ψ_2 opposite parity $\rightarrow \psi_1^2 x, \psi_2^2 x$ terms vanish

$$\langle x \rangle = \frac{1}{25} \int (-24 \psi_1 \psi_2 \sin(\omega t)) x dx = -\frac{24}{25} \sin(\omega t) \langle \psi_1 | x | \psi_2 \rangle$$

$$x = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-), \quad \langle \psi_1 | x | \psi_2 \rangle = \sqrt{\frac{\hbar}{2m\omega}} \langle \psi_1 | \sqrt{2} \psi_1 \rangle = \sqrt{\frac{\hbar}{m\omega}}$$

$$\langle x \rangle = -\frac{24}{25} \sqrt{\frac{\hbar}{m\omega}} \sin(\omega t)$$

$$m \frac{d\langle x \rangle}{dt} \rightarrow \langle p \rangle = -\frac{24}{25} \sqrt{\hbar m \omega} \cos(\omega t)$$

Ehrenfest's Theorem: $\frac{d\langle p \rangle}{dt} = \langle -\partial V / \partial x \rangle$

harmonic oscillator: $V(x) = \frac{1}{2} m \omega^2 x^2$

$$\rightarrow \frac{\partial V}{\partial x} = -m \omega^2 x$$

$$= -m \omega^2 \left(-\frac{24}{25} \sqrt{\frac{\hbar}{m\omega}} \sin(\omega t) \right)$$

$$= \frac{24}{25} \omega \sqrt{\hbar m \omega} \sin(\omega t)$$

$$\frac{d\langle p \rangle}{dt} = -\frac{24}{25} \sqrt{\hbar m \omega} (-\omega \sin(\omega t)) = \frac{24}{25} \omega \sqrt{\hbar m \omega} \sin(\omega t)$$

So $\frac{d\langle p \rangle}{dt} = \langle -\frac{\partial V}{\partial x} \rangle$. So Ehrenfest's thm holds

(d) energy, probability

$$E_1 = \frac{3}{2} \hbar \omega \quad E_2 = \frac{5}{2} \hbar \omega$$

↓

↓

$$\text{probability: } \left(\frac{3}{5}\right)^2 \\ = \frac{9}{25}$$

$$\text{probability: } \left| -\frac{4i}{5} \right|^2 \\ = \frac{16}{25}$$

3. $\Psi(x, 0) = \frac{A}{a^2 + x^2}$

(a) $\int_{-\infty}^{\infty} |\Psi(x, 0)|^2 dx = \int_{-\infty}^{\infty} \frac{A^2}{(a^2 + x^2)^2} dx = 1$

Let $x = a \tan \theta$, $dx = a \sec^2 \theta d\theta$, $x^2 + a^2 = a^2 \sec^2 \theta$.

$$\int_{-\pi/2}^{\pi/2} \frac{A^2}{(a^2(1 + \tan^2 \theta))^2} a \sec^2 \theta d\theta = \frac{A^2}{a^3} \int_{-\pi/2}^{\pi/2} \cos^2 \theta d\theta$$

$$\frac{A^2}{a^3} \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_{-\pi/2}^{\pi/2} = \frac{A^2}{a^3} \left(\frac{\pi}{4} - \left(-\frac{\pi}{4} \right) \right) = \frac{A^2 \pi}{2a^3} = 1$$

$$\rightarrow A^2 = \frac{2a^3}{\pi} \rightarrow A = \sqrt{\frac{2a^3}{\pi}}$$

(b) find $\phi(k)$

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{A}{a^2 + x^2} e^{-ikx} dx$$

$$= \frac{A}{\sqrt{2\pi}} \frac{\pi}{a} e^{-a|k|} = \frac{A\pi}{a\sqrt{2\pi}} e^{-a|k|} = \frac{\pi}{a\sqrt{2\pi}} \sqrt{\frac{2a^3}{\pi}} e^{-a|k|}$$

$$= \sqrt{a} e^{-|k|a}$$

(c) $\Psi(x, t)$

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{i(kx - \frac{\hbar k^2}{2m} t)} dk$$

$$\Rightarrow \Psi(x, t) = \sqrt{\frac{a}{2\pi}} \int_{-\infty}^{\infty} e^{-|k|a} e^{i(kx - \frac{\hbar k^2}{2m} t)} dk$$

(d)



4. $\Psi(x, t) = A e^{i(kx - \frac{\hbar k^2}{2m} t)}$

$$J(x, t) = \frac{i\hbar}{2m} \left(\Psi \frac{\partial \Psi^*}{\partial x} - \Psi^* \frac{\partial \Psi}{\partial x} \right)$$

$$\frac{\partial \Psi}{\partial x} = ik A e^{i(kx - \frac{\hbar k^2}{2m} t)} = ik \Psi$$

$$\frac{\partial \Psi^*}{\partial x} = -ik \Psi^*$$

$$\Rightarrow J(x, t) = \frac{i\hbar}{2m} \left(\Psi (-ik \Psi^*) - \Psi^* (ik \Psi) \right)$$

$$= \frac{i\hbar}{2m} (-2ik |\Psi|^2) = \frac{\hbar k}{m} |\Psi|^2 = \frac{\hbar k}{m} |A|^2$$

$$\therefore J(x, t) = \frac{\hbar k}{m} |A|^2$$

direction along k .